Coherence of Resilience Metrics for the Atlantic Meridional Overturning Circulation Across a Hierarchy of Models

Andreas Morr[1,2*] Reyk Börner[3] Sjoerd Terpstra[3]

Christian Kuehn[1]

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Abstract

The Atlantic Meridional Overturning Circulation (AMOC) is a critical component of the global climate system, and its potential collapse is one of the most consequential climate tipping events. Quantifying AMOC resilience, that is, its capacity to maintain the active circulation state under external perturbations, is therefore of high scientific and societal importance. Resilience is a multifaceted concept, encompassing notions such as local and global recovery times, basin volume, and the minimal perturbation required to trigger a collapse. Whether these distinct metrics yield consistent, and therefore trustworthy, signals across models of different complexity has not been systematically examined. Here, we address this question by analysing four models that span a hierarchy in physical complexity: (i) a 3-box salinity model, (ii) a two-dimensional latitude-depth Boussinesq model of the overturning circulation, (iii) the Earth system model of intermediate complexity CLIMBER-X, and (iv) the general circulation model PlaSim-LSG. Across the hierarchy, we compute model-specific implementations of four complementary resilience concepts: local resilience, convergence time, basin volume, and minimal

*Corresponding author: andreas.morr@tum.de

critical shock. Our results suggest that not all metrics are coherent across the model hierarchy, as exhibited by incompatible trends under increasing CO_2 forcing in each model. This points to deficiencies in model compatibility and challenges in applying these concepts in real-world systems. Several measures indicate an unexpected increase in system resilience under climate change. These findings have direct implications for the detectability of an imminent AMOC collapse from observational or model-based early-warning signals.

Keywords: Atlantic Meridional Overturning Circulation, tipping points, resilience, model hierarchy

1 Introduction

The Atlantic Meridional Overturning Circulation is one of the dominant features of the global ocean, transporting large amounts of warm, saline surface waters northward across the Atlantic and returning cold, deep North Atlantic water southward at depth. This large-scale transport pattern underpins the relatively mild climate of Western Europe and plays a central role in the global carbon and heat cycles. Palaeoclimate records document repeated rapid transitions between an active (AMOC-on) and a weakened or collapsed (AMOC-off) state during the Pleistocene, associated with Dansgaard–Oeschger events. Under continued anthropogenic forcing, rising sea surface temperatures, freshening of the North Atlantic from increased precipitation, and Greenland ice-sheet melt may push the AMOC towards a similar transition.

More concretely, in background climate configurations where at least two stable equilibria coexist (the on and off states), the system can be permanently tipped from one to the other by crossing a critical threshold leading to the annihilation of one of the formerly stable states (bifurcation-induced tipping) or by a sufficiently large state perturbation (shock-induced tipping) (**??**). Rate-induced tipping is another pathway, though it will not be relevant for this work. In the bifurcation-theory framework for AMOC tipping, the transition to the off state is often conceptualised as a saddle-node bifurcation (**?**). A shock-induced transition, on the other hand, can be understood as an exit of the system state from the basin of attraction associated with the AMOC on state. In both frameworks, the AMOC is investigated as an autonomous dynamical system that may be influenced by slow parameter changes or external perturbations, such as noise. This allows for the characterisation of the system in terms of several quantities, subsumed under the term "resilience" (**?**).

One important notion of resilience has been applied to the AMOC in the context of a bifurcation-induced transition. As the system approaches such a transition through a gradual change in the background climate, the stable on-state loses linear stability, a phenomenon known as critical slowing down (CSD) (**?**). CSD manifests in observable statistics: near the bifurcation, the dominant eigenvalue of the linearised system ap-

proaches zero, leading to increases in variance and autocorrelation of system fluctuations (**??**). These generic signatures have motivated a large body of research on early-warning indicators of tipping (**???**).

Yet CSD-based indicators capture only the system's reaction to small perturbations around the equilibrium, known as local resilience, quantifying how the on-state changes locally as the bifurcation is approached. A complementary perspective is provided by non-local resilience concepts, which characterise the size and geometry of the basin of attraction of the on-state and therefore also capture the system's capacity to recover from larger perturbations (**???**). A comprehensive assessment of the resilience of the AMOC on state should encompass many different notions. In this work, we consider four such choices. The following heuristic definitions explain the interpretation of each respective resilience measure. The concrete implementation and calculation in the context of each model will vary and are explained in Section 2.

1. **Local resilience**: capacity of a system to continuously absorb small perturbations to its equilibium on-state, such as noise and internal variability (**?**).

2. **Convergence time**: duration trajectories take to return to the on-state equilibrium after a large perturbation (**?**).

3. **Basin volume**: the volume fraction of a reference region of initial model conditions that converges to the on-state (**?**).

4. **Minimal critical shock**: the smallest instantaneous perturbation that moves the system out of the basin of attraction of the on-state (**??**).

In natural systems, controlled perturbation experiments are rarely feasible, making direct estimation of non-local resilience quantities from observations difficult. This is especially true for the AMOC, and numerical models, therefore, play an essential role. However, climate models span an enormous range of complexity, from analytically tractable conceptual models to full Earth system models (ESMs), and it is not obvious whether resilience metrics computed in simple models translate to their more complex counterparts,

which include many additional feedbacks and degrees of freedom (**??**). If metrics disagree across the hierarchy, it could mean that parts of the hierarchy do not expose them well. The extrapolation of findings on these metrics from models to the real-world system is then harder to justify. Several studies have examined individual resilience concepts in the context of the AMOC at specific levels of model complexity (**?????**), but a systematic, multi-metric comparison across a model hierarchy is lacking.

Our analysis combines four modelling studies that span many orders of magnitude of model complexity: (i) a conceptual 3-box salinity model (**??**), (ii) a latitude-depth Boussinesq partial differential equation model (**?**), (iii) the Earth system model of intermediate complexity CLIMBER-X (**?**), and (iv) the coupled general circulation model PlaSim-LSG (**?**). Section 2 describes each model and the resilience metrics computed for it. Section 3 reports the individual model's results and Section 4 synthesises them across the hierarchy. In Section 4, we discuss implications for research on the AMOC and on early-warning detection of potential tipping.

2 Methods

We compute a common set of resilience quantities for each model while respecting the different experimental possibilities at different levels of the hierarchy. More concretely, we analyse trajectories from perturbed initial conditions. The reduced state space in which perturbations take place is defined by North Atlantic and Tropical Atlantic salinity values. Salt is conserved by applying a compensating perturbation in the remaining parts of the world ocean. Figure 1 illustrates the model-specific perturbation and diagnostic regions, and their implementation is explained in the respective subsections below. Table 1 gives an overview of the concrete meaning and estimation routine for each resilience measure across the models. Initial conditions in the 3-box, Boussinesq, and CLIMBER-X experiments are probed within rectangles bounded by $\pm 2\,\mathrm{psu}$ perturbations relative to their model-specific reference configurations. This is done by drawing a large number of uniform samples from this region (3-box model) or by evenly sampling a grid of initial conditions (Boussinesq model and CLIMBER-X). PlaSim-LSG constitutes an exception:

we analyse existing edge-track and equilibrium runs from **?** using two salinity readouts in the North and Tropical Atlantic.

We repeat the resilience analysis across a global-warming forcing coordinate, as specified for each model below. This allows us to compare directional trends of resilience, which are more meaningful across models than absolute values expressed in model-specific units. The resilience of the AMOC on-state is the primary focus throughout, though all analyses could also be carried out for the off-state when present.

2.1 3-Box salinity model

We adopt the 5-box salinity model introduced by **?** and further reduced to 3 boxes by **?**. The original model partitions the Atlantic–Indo-Pacific basin into five boxes: North Atlantic (N), Tropical Atlantic (T), Southern Ocean (S), Indo-Pacific (IP), and a Deep box (B). In the implemented 3-box reduction, the North and Tropical Atlantic salinity anomalies, S_N and S_T, are the two dynamical state variables. The slow variables of Southern Ocean and Deep-box salinities are prescribed as constants by the parameter vector, while Indo-Pacific salinity is derived at each time from salt conservation. The exact model equations and parameters are given in Appendix A.

? calibrated two distinct parameter sets for the model: one for a pre-industrial climate ($p_{1\times}$) and one for a doubled-CO_2 climate ($p_{1\times}$) in the FAMOUS Earth System model. We construct a continuum of models through the full parameter vector as $p(r) = p_{1\times} + r(p_{2\times} - p_{1\times})$ and label it by $C(r) = 280\,\text{ppm} + 280\,\text{ppm}\,r$. Thus $t = 0$ and $t = 1$ correspond to 280 and 560 ppm, respectively, while $r > 1$ denotes linear extrapolation beyond the doubled-CO_2 parameter vector. The nominal spacing is $\Delta r = 0.05$, or 14 ppm, and we investigate the system from 280 through 896 ppm. We find that the on-state vanishes in a bifurcation when extrapolating the linear parameter ramp further, at around 900 ppm.

All numerical experiments and resilience calculations are performed using the Julia package Attractors.jl (**?**) within the DynamicalSystems.jl ecosystem. This computational framework enables efficient integration of large initial-condition ensembles and simultaneous tracking of multiple resilience metrics as a parameter changes (**?**). At each continua-

tion step, we probe the two-dimensional state space with 10 000 initial conditions sampled uniformly from a ± 2 psu rectangle centred on the on-state attractor. Local resilience is computed from the Jacobian of the right-hand side of the model equation. The resilience measured as $-\mathrm{Re}(\lambda_{\max})$ where $\lambda_{\max}$ is the eigenvalue with the largest real part. In Figure 2, we map out the local basin geometry on a regular grid and show several example trajectories in terms of AMOC strength and phase-space evolution.

2.2 Boussinesq latitude-depth model

The second model in our hierarchy is a latitude-depth Boussinesq model of the AMOC. The model evolves temperature, salinity, streamfunction, and vorticity on a 21×41 grid of discretised latitude and depth points. The one-dimensional sea surface is forced by cosine profiles of temperature and salinity flux, with lower temperatures and net freshwater input at high latitudes. This model is much less expensive than global circulation models but already contains spatially structured salinity fields and overturning, making it a useful intermediate step between the conceptual box model and the three-dimensional climate models.

We introduce a single external forcing parameter that mimics Arctic amplification: increasing it flattens the equator-to-pole surface temperature profile (Arctic amplification) and enhances the freshwater forcing profile (intensification of the hydrological cycle). For the exact mechanism of this parameter, its mapping to CO_2, and the exact model equations, see Appendix B. The perturbation experiments are performed across seven forcing values corresponding to CO_2 concentrations from 280 ppm to approximately 643 ppm. For each configuration, the model is first spun up twice under weak hemispheric salinity-forcing asymmetries to ensure convergence to the on- and off-states, respectively. Under the symmetric salinity forcing, for which an active and a reversed overturning state can coexist over part of the forcing range, we obtain the final AMOC on and off reference states by evolving the asymmetric-forcing equilibria until convergence. Based on the reference on-state, we initialise perturbed salinity trajectories. We then classify each perturbed run by the reference state it settles to, which in turn defines the basin-volume and

convergence-time metrics.

Initial-condition perturbations are applied to grid-aligned salinity boxes, and we utilise an affine conversion of the non-dimensional salinity units of the Boussinesq model to psu (for details see Appendix B). The North Atlantic box covers 40.5°–90°N, the Tropical Atlantic box covers 31.5°S–40.5°N, and the Southern compensation box covers the domain south of 31.5°S. All three extend through the upper approximately third of the water column. Cosine-tapered perturbation fields are normalised so that their mean over each hard box equals the requested amplitude. Perturbations are applied as salinity anomalies $(\Delta S_N, \Delta S_T)$ on a 20×20 grid (400 initial conditions), with both amplitudes spanning ± 2 psu relative to the on-state equilibrium; a compensating perturbation in the Southern box conserves total domain salinity.

For the basin-mapping ensemble, only the salinity field of the on-state equilibrium is carried over into the perturbed initial conditions; the temperature, vorticity, and streamfunction fields are set to zero and left to re-adjust to the perturbed salinity under the model dynamics. This isolates the salinity perturbation as the driver of the response. To estimate local linear stability, we run a second, smaller ensemble of eight experiments per forcing level. Each run has a salinity perturbation in the two-dimensional (North and Tropical Atlantic) salinity space. The eight perturbations are evenly arranged on a circle of radius $\varepsilon = 0.5$ psu around the on-state equilibrium. Unlike the basin-mapping ensemble, these perturbations are applied to the complete equilibrium state, retaining all temperature, vorticity, and streamfunction fields, so that the local recovery of the full system is reflected in the two-dimensional salinity projection. All trajectories are run under neutral symmetric salinity forcing for approximately 500 model years. After discarding an initial ≈50-year transient, a 2×2 discrete linear map M is fitted by least squares to trajectory pairs $(x_t,\, x_{t+\Delta t})$ sampled every ≈20 years, where $x = (\delta S_N,\, \delta S_T)$ denotes the box-salinity deviation from equilibrium. The continuous-time generator $A = \log(M)/\Delta t$ of the fitted map M yields eigenvalues whose least stable real part gives the local resilience $-\mathrm{Re}(\lambda_{\max})$.

Throughout, we report salinities in psu and the overturning strength in Sverdrups;

the affine salinity calibration and linear overturning conversion, both anchored on the pre-industrial equilibrium, are given in Appendix B. Time metrics are reported on a year-proxy scale equal to 100 times the model time unit. In Figure 3, we plot four exemplary trajectories in terms of the two box salinities and the AMOC strength at 280 ppm and approximately 488 ppm.

2.3 CLIMBER-X

CLIMBER-X (**?**) is a fast Earth system model of intermediate complexity (EMIC) that couples a 3-D ocean general circulation model, an energy-balance atmosphere, a thermodynamic sea-ice model, and land-surface components. The ocean component resolves the full three-dimensional salinity and temperature fields, including the main overturning circulation cells, at a resolution of approximately $5° \times 5°$ with 23 vertical levels. CLIMBER-X is computationally efficient enough to permit multi-millennial integrations, making it well-suited for exploring the state space through moderately sized perturbation ensembles.

We first equilibrate CLIMBER-X at 15 prescribed atmospheric CO_2 concentrations between 285 and 600 ppm. Each equilibrium run is integrated for up to 20 000 model years from the default pre-industrial restart. After equilibration, the trailing 500 years of the annual ocean time series are used to store reference values for AMOC strength at 26°N and for the salinity diagnostics in the geographical boxes illustrated in Fig. 1.

The North Atlantic box comprises Atlantic-basin grid cells from 35°–80°N and the Tropical Atlantic box comprises Atlantic-basin cells from 35°S–35°N. Both boxes cover the upper 105 m, corresponding to the top four ocean layers. The North and Tropical box perturbations use a two-grid-cell latitudinal taper and a two-layer vertical cosine taper. The tapered fields are normalised so that the volume-weighted mean anomaly over the hard target box remains exactly equal to the prescribed salinity perturbation. The salt added to the two Atlantic boxes is compensated uniformly over the rest of the world ocean, so that the global ocean salt content in the restart is conserved.

For our experiments, we apply instantaneous salinity perturbations to these boxes

directly to the ocean restart as salinity anomalies in psu, creating an ensemble of modified initial conditions. The North Atlantic and Tropical Atlantic perturbations each take the values $-2, -1.5, \ldots, 2$ psu, giving a 9×9 grid of 81 experiments per CO_2 level. Each perturbed initial condition is integrated forward for 10 000 model years. This model is known to exhibit an AMOC off-state under high CO_2 concentrations when large constant freshwater forcing in the North Atlantic is applied. In the present setup of pure initial salinity perturbations, the off-state is not reached. Instead, at higher CO_2 a second, distinct attractor with a somewhat stronger AMOC emerges. This occurs when sea ice in the northern hemisphere retreats and the deep-water convection driving the AMOC expands into areas farther north. We classify each grid trajectory by its final state using the same criterion as for the Bussinesq model: a run is assigned to the on-state if its final salinities lie within an $\varepsilon = 0.05$ psu ball of the unperturbed reference state, and to the "strong" state otherwise.

A further CO_2-dependence concerns the long equilibration run itself: the state on which it settles, and which serves as the unperturbed reference, changes branch with forcing. Up to 330 ppm it lies on the reference ("modern") branch, whereas from 345 ppm onward it lies on the stronger-AMOC ("strong") branch. The resilience measures obtained from the perturbation experiments are therefore associated with the modern state at 285–330 ppm and with the strong state at 345–600 ppm. The basin volume reported for CLIMBER-X (Table 1) is the share of the 81 grid trajectories that return to this reference equilibrium; it equals one wherever all perturbations recover and falls below one only near the modern-branch transition (at 315 and 330 ppm), where a fraction of perturbations settle on the strong state instead. From the same grid we obtain the minimal critical shock as the smallest salinity-plane distance from the reference equilibrium to a perturbed initial condition that does not return to its basin; it is a grid-resolution-limited upper bound and is defined only where at least one perturbation escapes, i.e. at 315 and 330 ppm, since at all other CO_2 levels the entire ± 2 psu grid recovers. Figure 4 shows four example trajectories starting from perturbed salinity initial conditions in the specified boxes, similar to the two figures above. For both CO_2 values, the equilibrium states are

the modern state.

To estimate local linear stability, we run a second, smaller ensemble of 9 experiments per CO_2 level. Eight runs apply isotropic salinity perturbations of magnitude $\varepsilon = 0.1$ psu again arranged on a circle in the two-dimensional (NA, Tropical) shallow-box salinity space, plus one unperturbed reference run. Each run is integrated for 500 model years. An equilibrium reference is obtained from the mean of the unperturbed run over years 10–150. A 2×2 discrete linear map is fitted by least squares to annual trajectory pairs (x_t, x_{t+1}) in the same fitting window. The continuous-time generator $A = \log(M)/(1\,\mathrm{yr})$ yields eigenvalues whose least stable real part gives the local resilience $-\mathrm{Re}(\lambda_{\max})$.

2.4 PlaSim-LSG

The last component of the hierarchy is the coupled climate model of intermediate complexity PlaSim-LSG, comprising a spectral primitive-equation atmosphere (T21, $\approx 5.6^\circ \times 5.6^\circ$, 10 vertical levels) and the Large Scale Geostrophic (LSG) ocean (**?**), a full 3-D ocean circulation model at $3.5^\circ \times 3.5^\circ$ resolution with 22 vertical levels, coupled to a thermodynamic sea-ice scheme and a hydrological cycle. With approximately 10^5 degrees of freedom and a throughput of roughly 700 model years per CPU day, PlaSim-LSG is considerably more complex than CLIMBER-X. In particular, it resolves atmospheric variability more faithfully and thereby yields seemingly chaotic ocean dynamics.

At 285 ppm (pre-industrial) and 360 ppm (current) CO_2, PlaSim-LSG exhibits a bistable AMOC with an on-state of ≈ 15 Sv and an off-state (deep convection in Subpolar Gyre ceases) of $\approx 1 - 2$ Sv, both confirmed by long unforced simulations. **?** applied the edge-tracking algorithm (**?**) to locate the edge state (Melancholia state) of the AMOC in PlaSim-LSG at 285 ppm and 360 ppm. The algorithm iteratively bisects pairs of trajectories converging to opposite attractors and tracks the resulting pseudotrajectory along the basin boundary until it converges to the edge state. The edge state is a chaotic saddle characterized by persistent centennial AMOC oscillations (≈ 120 yr period, amplitude 3–10 Sv), driven by atmosphere–ice–ocean interactions in the North Atlantic. A complete description of the model configuration, its AMOC bistability, and the edge-tracking

procedure is given in **?**, to which we refer the reader for full methodological details.

From the final iterations of edge tracking at each CO_2 level, we extract pairs of edge tracks: one converging to the on-state and one converging to the off-state, both initialised from near-boundary conditions that closely shadow the edge state before diverging. These trajectories provide a natural set of initial conditions from which resilience-relevant transient behaviour can be quantified: they start close to the basin boundary and thus represent challenging recovery paths for the on-state. In total, we analyse 37 paired edge-track files at 285 ppm and 38 at 360 ppm. Dedicated long equilibrium simulations of the AMOC-on, AMOC-off, and edge states at each CO_2 level provide the attractor clouds against which these transients are evaluated.

We reduce the high-dimensional PlaSim-LSG salinity output to the salinity of two geographical boxes. The North Atlantic readout is the mean salinity from 35°–80°N and 0–1000 m depth, and the Tropical Atlantic readout covers 35°S–35°N and 0–500 m depth. Dedicated on-, off-, and edge-state equilibria trajectories determine the means associated with these states in this two-dimensional salinity plane. The covariance matrices of the scattered trajectory points are computed to obtain an approximate Gaussian distribution of data for each state. Convergence time is measured from the last visit to the edge-state 4σ Gaussian ellipse to the first entry into the target-attractor ellipse, and the critical-shock proxy is the gap between the edge-state and on-state ellipses. Local resilience is quantified as the inverse width of the on-state fluctuation cloud, $[2000\sqrt{\lambda_{\max}(C)}]^{-1}$, where $\lambda_{\max}(C)$ is the largest eigenvalue of the on-state equilibrium covariance C; a broader cloud, i.e. larger salinity fluctuations around the on-state, corresponds to lower local resilience.

3 Results

Increasing CO_2 does not produce a common change in AMOC strength or resilience across the hierarchy (Fig. 6). The minimal-critical-shock estimates generally rise over the intervals where finite values are available in each model, but this does not consistently coincide with faster recovery or a larger sampled on-state basin.

The three-box model is the only member whose reference AMOC weakens throughout

the sampled range, from 15.1 Sv at 280 ppm to 6.8 Sv at 896 ppm, before the on-state is lost near 900 ppm along the extrapolated parameter path (Fig. 6a). The Boussinesq reference diagnostic instead increases from 18.0 to 20.8 Sv over 280–643 ppm. These values describe the weakly asymmetric spin-up; perturbation outcomes are evaluated under symmetric forcing. CLIMBER-X strengthens from 19.7 Sv at 285 ppm to 21.7 Sv at 330 ppm on the modern branch. From 345 ppm onward, the reference equilibrium lies on the strong branch, reaching approximately 24 Sv above 400 ppm. PlaSim-LSG changes little between its two available climates, with mean on-state strength increasing from 14.72 Sv at 285 ppm to 15.03 Sv at 360 ppm.

The perturbed trajectories distinguish temporary weakening from a change of attractor. In the three-box model, the illustrated perturbation that increases North Atlantic salinity and decreases Tropical Atlantic salinity leads to collapse at 280 ppm but recovery at 448 ppm, following a pronounced weakening and damped overshoot (Fig. 2). Boussinesq trajectories can spend more than a millennium with weak overturning before recovering, including the adjustment of the initially reset temperature and circulation fields (Fig. 3). Both simpler models admit transitions to reversed overturning. CLIMBER-X perturbations instead either recover or reach a distinct strong state, whose AMOC at 330 ppm is only about 0.4 Sv stronger than the modern state; none reaches an off-state (Fig. 4). In PlaSim-LSG, trajectories departing from the edge state either recover, commonly overshooting the on-state mean, or approach a weak residual circulation of approximately 1–2 Sv (Fig. 5).

Local resilience decreases smoothly in both simpler models despite their opposing trends in AMOC strength (Fig. 6b). The three-box rate falls from $0.0100\,\mathrm{yr}^{-1}$ at 280 ppm to $0.00153\,\mathrm{yr}^{-1}$ at 882 ppm, corresponding to a characteristic return-time increase from 100 to 654 model years. The Boussinesq decrease is more modest, equivalent to a return-time increase from 65 to 78 years on the adopted year-proxy scale. PlaSim-LSG gives a small change in the opposite direction: its covariance-based proxy increases by approximately 3%, reflecting a slightly narrower on-state salinity fluctuation cloud rather than an estimated decay rate.

CLIMBER-X local resilience varies non-monotonically. On the modern branch, it falls from $0.0495\,\mathrm{yr}^{-1}$ at 285 ppm to $0.0263\,\mathrm{yr}^{-1}$ at 315 ppm, then rises to $0.0326\,\mathrm{yr}^{-1}$ at 330 ppm. On the strong branch, maxima near 345 and 430 ppm are followed by a minimum of $0.0167\,\mathrm{yr}^{-1}$ at 510 ppm and recovery to $0.0309\,\mathrm{yr}^{-1}$ at 600 ppm. These substantial variations accompany comparatively small changes in equilibrium overturning along the upper branch.

Mean recovery from finite perturbations slows more strongly than local recovery in the two simpler models (Fig. 6c). Three-box convergence time increases from 136 model years at 280 ppm to 272 at 560 ppm and 1595 at 882 ppm. The Boussinesq mean rises from 616 to 1945 years over 280–643 ppm on the year-proxy scale, more than tripling while the local return time increases by only about 20%. PlaSim-LSG edge-to-on transit time also increases, from 105 to 151 years. Its longer recovery thus accompanies a slightly narrower equilibrium fluctuation cloud.

CLIMBER-X mean convergence times have pronounced peaks. At 315 ppm, the mean reaches 923 years while the median is only 87 years; at 330 ppm, the mean falls to 147 years. On the strong branch, means peak near 1070–1080 years at 490–510 ppm, where local resilience is also low, before falling to approximately 580 years at 600 ppm. Medians remain between 63 and 102 years across that branch. The large means therefore reflect a subset of slowly returning trajectories rather than a comparable slowing of the typical response. At 315 and 330 ppm, these statistics exclude trajectories reaching the alternative strong state.

Despite slower recovery, a larger fraction of sampled perturbations returns to the on-state in the two simpler models (Fig. 6d). The three-box basin fraction rises from 0.516 to approximately 0.80, and the Boussinesq fraction from 0.710 to 0.788. Their minimal-critical-shock estimates increase as well (Fig. 6e): from 0.53 to 1.26 psu between 280 and 882 ppm in the three-box model, and from 0.81 to 1.05 psu over 280–643 ppm in the Boussinesq model. At the final three-box step, 896 ppm, basin fraction and critical shock decline slightly to 0.798 and 1.21 psu, while mean convergence time exceeds 3000 years. Thus, the loss of local resilience near the end of the continuation is accompanied by only

a small decrease in these geometric measures. The fractions and escape distances describe the prescribed finite samples, with invalid Boussinesq runs excluded.

In CLIMBER-X, basin fraction and minimal critical shock move in opposite directions between 315 and 330 ppm. All 81 perturbations return to the modern state at 285 and 300 ppm, but only 62 return at 315 ppm and 36 at 330 ppm, giving fractions of 0.765 and 0.444. The remaining trajectories reach the strong state. The minimal critical shock nevertheless increases between these two levels: fewer sampled perturbations recover, yet the nearest sampled escape lies farther from the modern equilibrium. Over the same interval, local resilience rises and conditional mean convergence time falls, also indicating greater resilience by those measures. From 345 ppm onward, every grid perturbation returns to the strong reference state. Finite critical-shock estimates are therefore obtained only at 315 and 330 ppm; elsewhere no escape is resolved within the sampled grid. These shocks cause transitions between two active circulation states, not AMOC collapse.

The PlaSim-LSG critical-shock proxy also increases, with the gap between the on-state and edge-state 4σ ellipses widening from 0.050 to 0.110 psu between 285 and 360 ppm. The larger separation accompanies longer edge-to-on transit times. For the off-state, both changes reverse: its gap to the edge-state ellipse decreases from 0.204 to 0.093 psu, and transit time falls from 116 to 75 years. No basin-volume comparison is available for PlaSim-LSG because the edge tracks do not sample a reference region of initial conditions.

4 Discussion

[TODO: Coherence of resilience metrics across the hierarchy, Global versus local metrics, Implications for early-warning detection, Limitations and caveats]

A Specifics on the 3-Box salinity model

The model tracks North Atlantic and Tropical Atlantic salinity anomalies S_N and S_T in the internal convention $S_i = 100(S_i^{\mathrm{dim}} - S_0)$, where $S_0 = 0.035$. In the figures, these values are converted into practical salinity as $S_i^{\mathrm{plot}} = 35 + 10S_i$ psu. Southern Ocean

salinity S_S and Deep-box salinity S_B are prescribed and converted to the same anomaly convention. Indo-Pacific salinity is instead diagnosed from total salt content C at every model evaluation:

$$S_{IP} = \frac{100}{V_{IP}}\left[C - \frac{V_N S_N + V_T S_T + V_S S_S + V_B S_B}{100} - S_0(V_N + V_T + V_S + V_{IP} + V_B)\right]. \quad (1)$$

The hosing-adjusted freshwater fluxes are $\widetilde{F}_N = F_N + 0.1311 \times 10^6 H$ and $\widetilde{F}_T = F_T + 0.6961 \times 10^6 H$. The continuation used here has $H = 0$ in both endpoint parameter vectors. The implemented governing equations are

$$\frac{\mathrm{d}S_N}{\mathrm{d}t} = \begin{cases} \dfrac{Y}{V_N}\left[\dfrac{(q + K_N)(S_T - S_N)}{100} - \widetilde{F}_N S_0\right], & q \geq 0, \\ \dfrac{Y}{V_N}\left[-\dfrac{q(S_B - S_N)}{100} + \dfrac{K_N(S_T - S_N)}{100} - \widetilde{F}_N S_0\right], & q < 0, \end{cases} \quad (2)$$

$$\frac{\mathrm{d}S_T}{\mathrm{d}t} = \begin{cases} \dfrac{Y}{V_T}\Big[\dfrac{q(\gamma S_S + (1-\gamma)S_{IP} - S_T)}{100} + \dfrac{K_S(S_S - S_T)}{100} \\ \qquad + \dfrac{K_N(S_N - S_T)}{100} - \widetilde{F}_T S_0\Big], & q \geq 0, \\ \dfrac{Y}{V_T}\Big[-\dfrac{q(S_N - S_T)}{100} + \dfrac{K_S(S_S - S_T)}{100} \\ \qquad + \dfrac{K_N(S_N - S_T)}{100} - \widetilde{F}_T S_0\Big], & q < 0, \end{cases} \quad (3)$$

$$q = \frac{\lambda\left[\alpha(T_S - T_0) + \beta(S_N - S_S)/100\right]}{1 + \lambda\alpha\mu}. \quad (4)$$

Here V_i are box volumes, K_N and K_S are mixing coefficients, F_N and F_T are freshwater fluxes, and Y is the model's time-scaling parameter. The routing parameter γ sets the fraction of positive overturning return flow passing through the Southern Ocean rather than the Indo-Pacific. The overturning flow q is reported in $\mathrm{m^3\,s^{-1}}$ and converted to Sverdrups by division by 10^6. Its dependence on flow direction makes the right-hand side discontinuous at $q = 0$.

The parameter vectors are taken from **?** and **?**, who calibrated pre-industrial ($1\times CO_2$) and doubled-CO_2 ($2\times CO_2$) configurations against the FAMOUS Earth System model. We linearly interpolate and extrapolate the full vectors, including changing box volumes, freshwater fluxes, temperatures, mixing coefficients, hydraulic and routing parameters,

and total salt content, as described in Section 2.1.

B Specifics on the Boussinesq model

Physical setup

The model represents a two-dimensional meridional cross-section of the ocean at non-dimensional latitude $x \in [0, 5]$ and depth $z \in [0, 1]$, where $x = 0$ corresponds to the South Pole, $x = 2.5$ to the equator, and $x = 5$ to the North Pole; $z = 0$ is the ocean floor and $z = 1$ is the surface (**?**). The zonally averaged flow is spatially two-dimensional and incompressible under the Boussinesq approximation, in which density variations are retained only in the buoyancy term. The velocity field is derived from a streamfunction $\psi(x, z, t)$ via

$$u = \frac{\partial \psi}{\partial z}, \qquad w = -\frac{\partial \psi}{\partial x}, \tag{5}$$

where u is the horizontal (meridional) velocity and w is the vertical velocity. The vorticity is $\omega = -\nabla^2 \psi$. The non-dimensional base parameters and numerical resolution are given in Table 2; the salinity asymmetry is changed between spin-up and perturbed integrations as described below.

Governing equations

Temperature T, salinity S, and vorticity ω evolve according to coupled advection–diffusion equations with surface forcing:

$$\frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x} + w\frac{\partial T}{\partial z} = \nabla^2 T + \frac{h(z)}{\tau_T}\big(T_s^\gamma(x) - T\big), \tag{6}$$

$$\frac{\partial S}{\partial t} + u\frac{\partial S}{\partial x} + w\frac{\partial S}{\partial z} = \frac{1}{\mathrm{Le}}\nabla^2 S + \frac{h(z)}{\tau_S}\, S_s^\gamma(x), \tag{7}$$

$$\frac{\partial \omega}{\partial t} + u\frac{\partial \omega}{\partial x} + w\frac{\partial \omega}{\partial z} = \mathrm{Pr}\,\nabla^2 \omega + \mathrm{Pr}\cdot\mathrm{Ra}\,\frac{\partial}{\partial x}(T - S), \tag{8}$$

$$\nabla^2 \psi = -\omega. \tag{9}$$

The surface boundary-layer profile $h(z) = \exp(-(1-z)/\delta)$ confines the surface forcing to the uppermost ocean. The temperature equation relaxes towards the prescribed surface profile with time scale τ_T; a smaller τ_T enforces temperature more strongly towards T_s. For salinity, a fixed flux formulation is used instead of a relaxation, which is physically appropriate since ocean salinity is set by freshwater exchange with the atmosphere rather than by direct relaxation to a reference profile. A global salinity correction is applied at each time step to conserve domain-mean salinity exactly.

The surface temperature profile is

$$T_s(x) = \tfrac{1}{2}\left[\cos\left(\tfrac{2\pi x}{a} - \pi\right) + 1\right], \tag{10}$$

giving $T_s = 0$ at both poles and $T_s = 1$ at the equator. This equator-to-pole contrast drives the horizontal density gradient that sustains the overturning. The surface salinity flux is

$$S_s(x) = 3.5\cos\left(\tfrac{2\pi x}{a} - \pi\right) - \beta_s \sin\left(\tfrac{\pi x}{a} - \tfrac{\pi}{2}\right), \tag{11}$$

where the dominant cosine term represents the atmospheric evaporation–precipitation cycle (net evaporation in the tropics, net precipitation at high latitudes) and the β_s term introduces a weak hemispheric asymmetry that adds a slight freshwater excess in the northern high latitudes relative to the south.

The vorticity equation (8) is obtained by taking the curl of the momentum equation. The baroclinic production term $\mathrm{Pr}\cdot\mathrm{Ra}\,\partial_x(T-S)$ encodes the linearised equation of state under the Boussinesq approximation: density anomalies $\rho \propto -(T-S)$ create horizontal density contrasts that drive overturning vorticity. The product $\mathrm{Pr}\cdot\mathrm{Ra} = 4\times10^4$ governs the balance between advective and diffusive vorticity transport and is the primary control on whether a sustained circulation develops.

Boundary conditions and numerical discretisation

No-flux ($\partial_n T = \partial_n S = 0$) and no-slip conditions are imposed on all lateral boundaries. The streamfunction satisfies $\psi = 0$ on all walls, consistent with a closed domain with

no net overturning flux through any boundary. The depth coordinate is discretised on a non-uniform grid that concentrates points near the surface and the ocean floor via the mapping

$$z_j = \tfrac{1}{2} + \frac{\tanh\big(3(j/N - \tfrac{1}{2})\big)}{2\tanh(3/2)}, \qquad j = 0, \ldots, N, \tag{12}$$

and the latitude coordinate uses a uniform grid. Spatial derivatives are approximated by second-order finite differences. Equations (6)–(8) are integrated using an implicit-explicit (IMEX) scheme with time step $\Delta t = 10^{-3}$: diffusion and surface-restoring terms are treated implicitly to avoid the diffusive CFL restriction, while advection terms are treated explicitly. The implicit step at each time leads to a separable Sylvester equation that is solved with the Bartels–Stewart algorithm.

Arctic amplification and CO_2 mapping

To sweep the model across different climate states, the surface temperature forcing is modified to mimic polar amplification by blending the standard profile towards its spatial mean:

$$T_s^{\gamma}(x) = (1 - \gamma)\, T_s(x) + \gamma \overline{T_s}, \tag{13}$$

where $\overline{T_s}$ is the mean of the unmodified temperature profile on the model grid and $\gamma \in [0, 1]$ measures the degree of Arctic amplification. At $\gamma = 0$ the forcing is unchanged (pre-industrial); increasing γ reduces the equator-to-pole temperature gradient, weakening the baroclinic torque in (8) and thereby reducing the thermal driving of the overturning.

The same parameter also intensifies the hydrological forcing: under a warming climate the evaporation–precipitation contrast strengthens ("wet-gets-wetter"), which we represent by scaling the symmetric component of the surface salinity flux (11) while leaving the hemispheric asymmetry unchanged,

$$S_s^{\gamma}(x) = (1 + \alpha_s \gamma)\, 3.5 \cos\!\big(\tfrac{2\pi x}{a} - \pi\big) - \beta_s \sin\!\big(\tfrac{\pi x}{a} - \tfrac{\pi}{2}\big)\,, \tag{14}$$

with hydrological-intensification coefficient $\alpha_s = 1$, so that the evaporation–precipitation amplitude would double at $\gamma = 1$ and increases by up to 6% across the sweep ($\gamma \leq 0.06$).

Increasing γ thus simultaneously weakens the thermal driving and strengthens the salinity forcing of the overturning.

A physically grounded mapping between γ and atmospheric CO_2 concentration C is derived from the logarithmic forcing formula and an estimate of polar ocean-surface amplification:

$$\gamma(C) = \frac{(\alpha_{\mathrm{sst}} - 1)\,\mathrm{ECS}}{\Delta T_0} \log_2\left(\frac{C}{C_0}\right) = 0.05 \log_2\left(\frac{C}{280\,\mathrm{ppm}}\right). \tag{15}$$

We adopt the IPCC AR6 best-estimate equilibrium climate sensitivity ECS = 3.0 °C per CO_2 doubling (**?**), a polar ocean-surface amplification factor $\alpha_{\mathrm{sst}} = 1.5$, and a pre-industrial Atlantic equator-to-pole SST difference $\Delta T_0 = 30\,^{\circ}\mathrm{C}$ (approximate Levitus climatology value). The Arctic sea-surface temperature response, which is the quantity relevant for the circulation-driving temperature field, is far weaker than the frequently quoted surface-*air* amplification: it is strongly damped by sea-ice insulation and local ocean heat uptake, and is of order 1.5 times the global-mean SST warming (**?**). These choices give $\gamma = 0$ at pre-industrial CO_2 ($C_0 = 280\,\mathrm{ppm}$) and $\gamma = 0.05$ at doubled CO_2 (560 ppm). The parameter sweep covers seven uniformly spaced values of $\gamma \in [0, 0.06]$ (steps of 0.01), corresponding to CO_2 concentrations from 280 ppm to approximately 643 ppm.

Perturbation geometry and physical scaling

Each salinity box is a finite set of grid cells whose lateral and lower boundaries are snapped to grid-cell faces. The North Atlantic box covers $x \in [3.625, 5]$ (approximately 40.5°–90°N), the Tropical Atlantic box covers $x \in [1.625, 3.625]$ (approximately 31.5°S–40.5°N), and the Southern compensation box covers the domain south of $x = 1.625$. All boxes include cells with $z \geq 1 - 0.28449$, corresponding to the top 15 levels and a lower face near 284 m on the 1000 m plotting scale. A cosine taper is applied at interior box edges while domain-wall edges remain sharp. Each tapered field is normalised so that its mean over the corresponding hard membership mask equals one; adding amplitude

ΔS therefore changes the box-mean salinity by exactly ΔS. North Atlantic and Tropical Atlantic perturbations are compensated in the Southern box to preserve the model's domain cell-mean salinity.

A spin-up with weak hemispheric asymmetry $\beta_s = 0.1$ seeds the symmetric-forcing ($\beta_s = 0$) AMOC-on equilibrium. A second $\beta_s = -0.1$ spin-up similarly seeds the reversed-AMOC equilibrium. At the start of each ensemble, both spin-up states are released under $\beta_s = 0$ dynamics and integrated until they settle. Basin-sweep initial conditions retain only the perturbed salinity field of the $\beta_s = 0.1$ spin-up; temperature, vorticity, and streamfunction are set to zero. A trajectory is stopped when its box salinities enter the $\varepsilon = 0.05$ psu neighbourhood of either startup equilibrium. Valid final streamfunction values are clustered independently, the strongest cluster is designated AMOC-on, and basin volume is the fraction assigned to that cluster. Mean convergence time is then evaluated for those AMOC-on trajectories relative to the median final box salinities of the cluster.

Model salinities are mapped to practical salinity units by the affine calibration $S\,[\,\mathrm{psu}] = 10.72735\,s + 25.45555$, fixed by requiring the North Atlantic and Tropical boxes of the pre-industrial ($\gamma = 0$, $\beta_s = 0.1$) equilibrium to match 35.0 and 36.5 psu; the ± 2 psu perturbation grid therefore corresponds to approximately ± 0.18644 model units. Likewise, the overturning streamfunction is converted to Sverdrups by anchoring the pre-industrial equilibrium value $\psi_0 \approx 1.517$ model units to 18 Sv, giving a factor of approximately $11.864\,\mathrm{Sv}/\psi_{\mathrm{model}}$. Convergence and characteristic return times are multiplied by 100 to obtain the year-proxy scale used in the synthesis, and local resilience is divided by 100 to obtain inverse year-proxy units.

C Specifics on the CLIMBER-X model

D Specifics on the PlaSim-LSG model

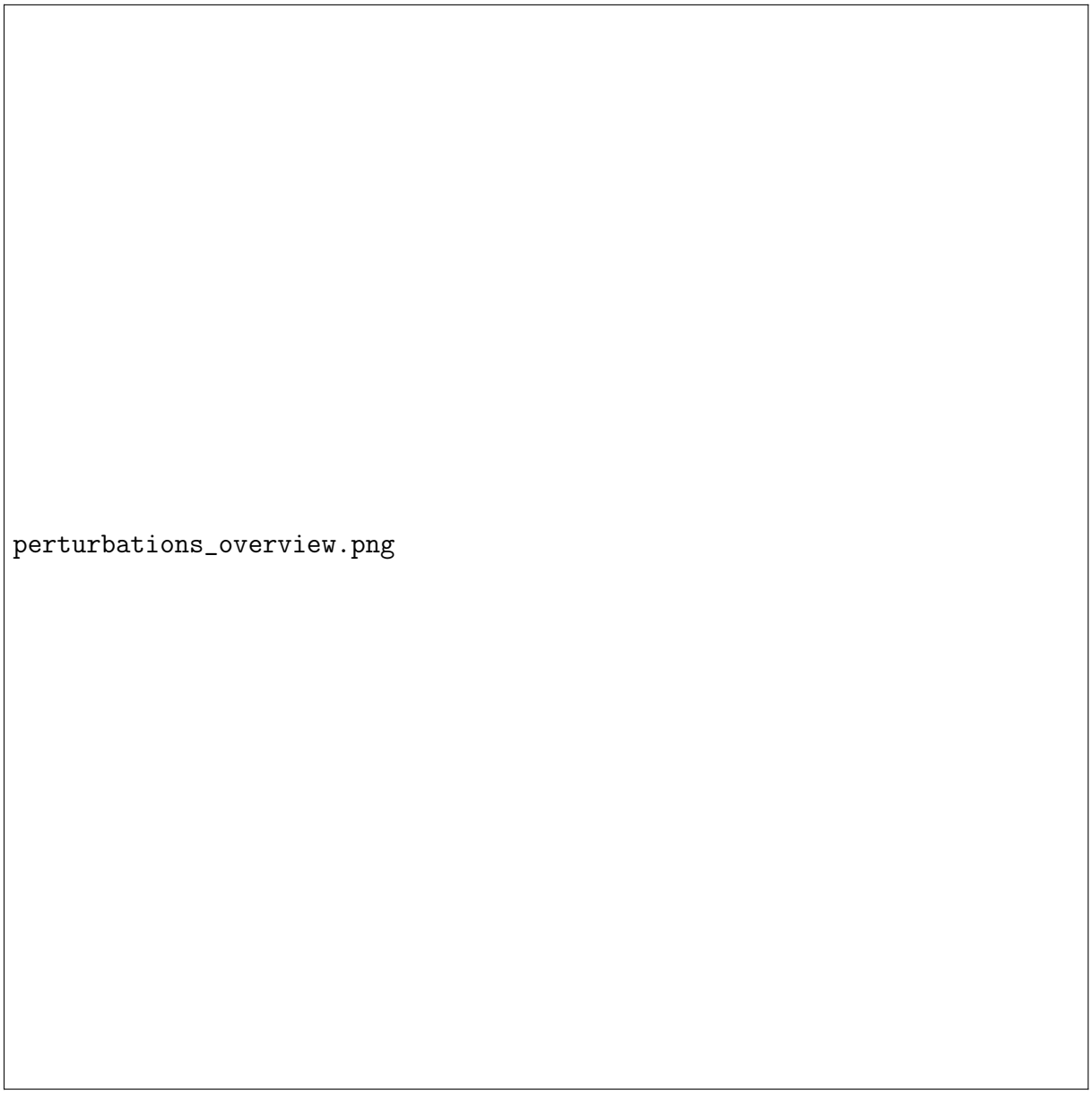


Figure 1: **Perturbation and diagnostic geometry across the model hierarchy.** This overview compares the geographic salinity regions used as perturbation coordinates (shaded regions) in the Boussinesq and CLIMBER-X experiments and as diagnostic read-outs (dashed boxes) in the Boussinesq, CLIMBER-X, and PlaSim-LSG analysis. The box geometry associated with the 3-box model is approximately reproduced from the model conception in **?**.

Table 1: **Overview of resilience-metric implementations across the model hierarchy.**

	3-Box model	**Boussinesq**	**CLIMBER-X**	**PlaSim-LSG**
Local resilience	$-\mathrm{Re}(\lambda_{\max})$: negative real part of the least stable eigenvalue of the Jacobian at the on-state fixed point, evaluated via forward-mode automatic differentiation (**?**).	$-\mathrm{Re}(\lambda_{\max})$: negative real part of the least stable eigenvalue fitted from a 2×2 discrete linear map via isotropic salinity perturbations.	$-\mathrm{Re}(\lambda_{\max})$: negative real part of the least stable eigenvalue fitted from a 2×2 discrete linear map via isotropic salinity perturbations.	Inverse of the long axis of the on-state fluctuation cloud ellipse.
Convergence time	Mean return time of on-state trajectories among 10 000 uniform random initial conditions in a ±2 psu box; convergence at $\varepsilon = 0.05$ psu proximity.	Mean return time of valid on-state trajectories on a 20×20 grid over ±2 psu; convergence at $\varepsilon = 0.05$ psu proximity.	Mean return time of completed trajectories on a 9×9 grid over ±2 psu; convergence defined by permanent entry into an $\varepsilon = 0.05$ psu ball.	Mean transit time from the last edge-state 4σ ellipse visit to the first on-state 4σ ellipse entry.
Basin volume	Fraction of 10 000 random initial conditions in a ±2 psu box centred on the on state that converge to the on state.	Fraction of valid initial conditions on the 20×20 perturbation grid converging to the on state.	Fraction of the 9×9 grid initial conditions over ±2 psu that converge back to the reference equilibrium (permanent entry into an $\varepsilon = 0.05$ psu ball).	Not estimated because the available edge tracks do not sample a reference region of state space.
Minimal critical shock	Minimum salinity-plane distance from the on state to a sampled initial condition not converging to the on state.	Minimum salinity-plane distance from the on state to a valid sampled initial condition not converging to the on state.	Minimum salinity-plane distance from the on-state to a sampled initial condition not converging to the on state.	Gap between the on-state and edge-state 4σ Gaussian ellipses in the deep-box salinity plane.



Figure 2: **Three-box model response to salinity perturbations.** Top panels (a, b) show the evolution of the AMOC strength for four representative initial conditions, with dashed horizontal lines marking the on- and off-state equilibria. Bottom panels (c, d) show the same trajectories in the two-dimensional salinity state space (the two state variables), overlaid on the sampled basins of attraction; circles represent 0.05 psu neighbourhoods around the on- and off-state fixed points (blue and orange, respectively). The left and right columns correspond to 280 and 448 ppm model specifications, respectively.



Figure 3: **Boussinesq model response to salinity perturbations.** Top panels (a, b) show the evolution of the AMOC strength for four representative initial conditions, with dashed horizontal lines marking the on- and off-state equilibria. Bottom panels (c, d) show the same trajectories in North Atlantic and Tropical Atlantic box-mean salinity, overlaid on the sampled basins of attraction. White areas represent initial conditions that timed out or encountered negative salinity and are excluded from metrics. Circles represent 0.05 psu neighbourhoods around the on- and off-state fixed points (blue and orange, respectively). The left and right columns correspond to 280 and 488 ppm model specifications, respectively.



Figure 4: **CLIMBER-X model response to salinity perturbations.** Top panels (a, b) show the AMOC strength at 26°N along four example trajectories from perturbed salinity initial conditions. Bottom panels (c, d) show the same trajectories in the two-dimensional Atlantic-box salinity plane, with axes spanning exactly the ±2 psu perturbation window. The shaded background maps the basin of attraction of every initial condition on the full 9×9 grid: cells whose trajectory returns to the reference on-state (final salinity within $\varepsilon = 0.05$ psu) are blue, and cells that settle on the distinct stronger-AMOC ("strong") state are purple. Circles mark the $\varepsilon = 0.05$ psu neighbourhoods of the on-state (blue, at the unperturbed equilibrium) and of the strong state (purple, at the mean endpoint of the non-converged trajectories). The two columns show prescribed atmospheric CO_2 concentrations of 285 ppm and 330 ppm: at 285 ppm all 81 initial conditions return to the on-state, whereas at 330 ppm 45 of the 81 converge to the strong state, which lies only ≈0.05–0.08 psu from the on-state but exhibits an AMOC about 0.4 Sv stronger.

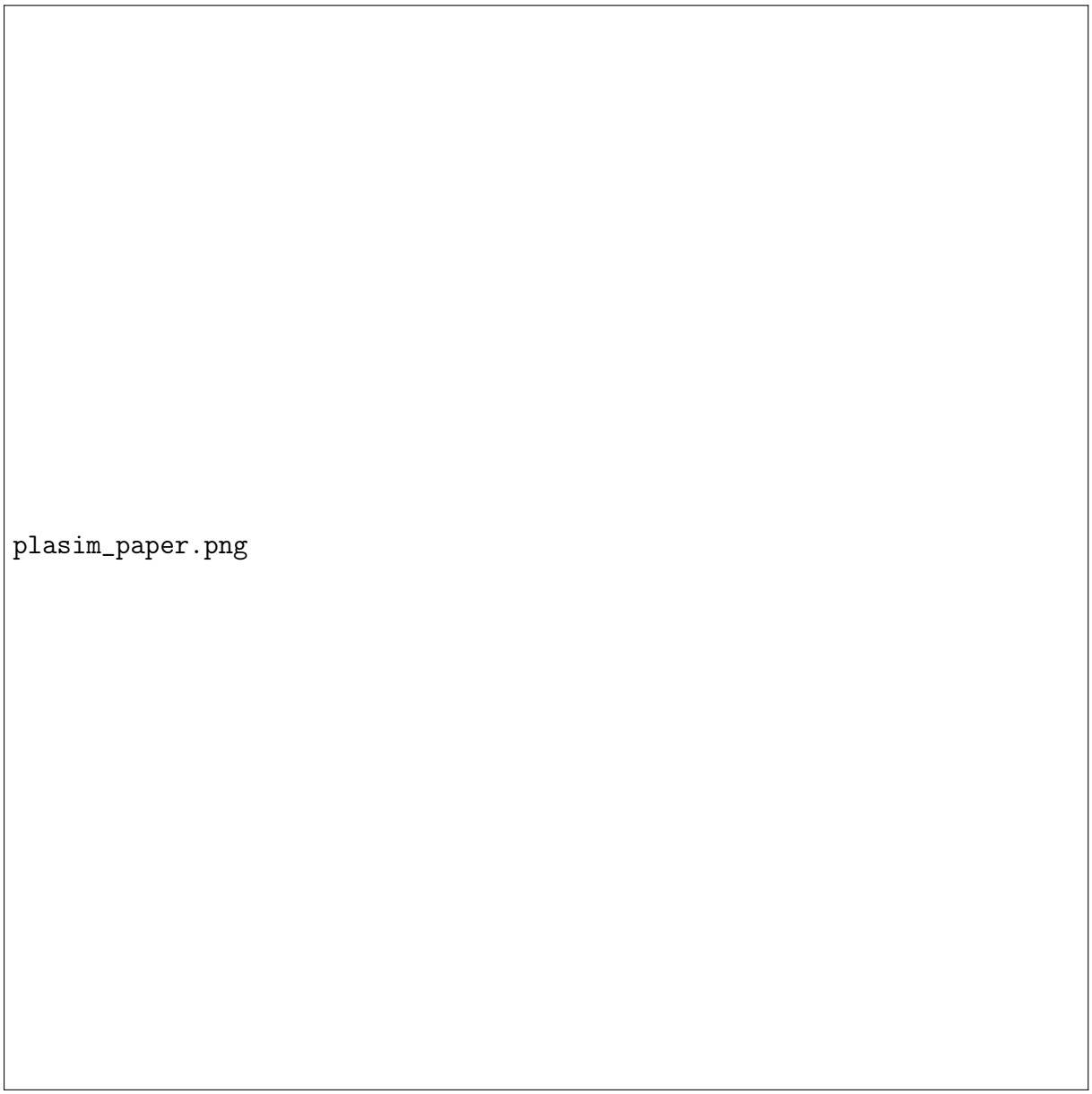


Figure 5: **PlaSim-LSG edge-state trajectories and covariance ellipses of attractors.** Top panels (a, b) show the AMOC strength readout along converged edge-track trajectories. Bottom panels (c, d) show the same transients in the North and Tropical Atlantic salinity coordinates. Ellipses denote the fitted 4σ Gaussian regions to equilibrium runs of the three states, used to define convergence times and edge-to-attractor distances. The two columns show the experiment results at prescribed atmospheric CO_2 concentrations of 285 ppm and 360 ppm, respectively.

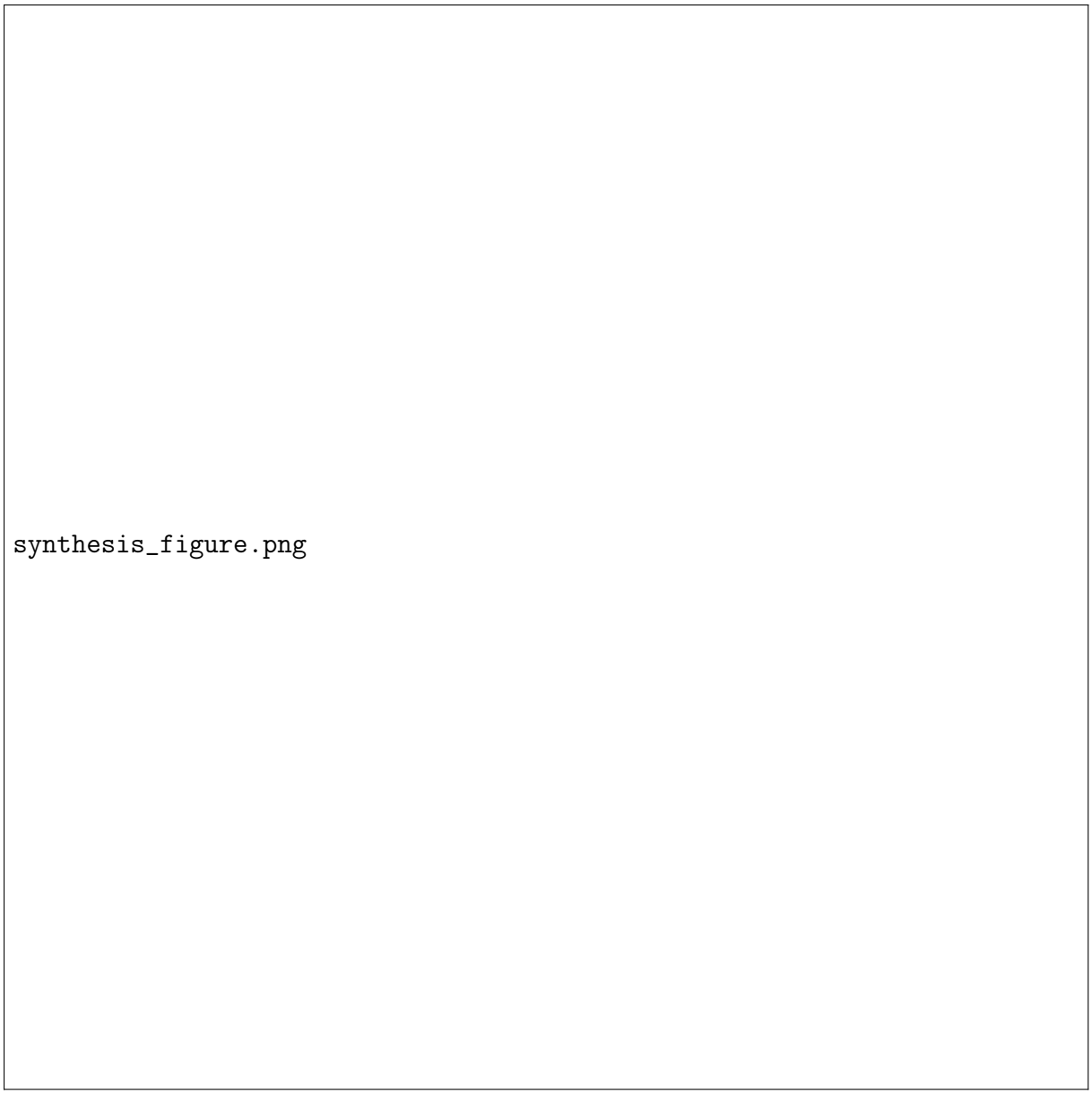


Figure 6: **Cross-model synthesis of AMOC resilience metrics.** (a) Reference AMOC strength, (b) local resilience, (c) mean convergence time, (d) sampled basin fraction, and (e) minimal critical shock. Definitions and model-specific scalings are given in Table 1 and the Methods. Larger values in (b), (d), and (e), and smaller values in (c), indicate greater resilience. CLIMBER-X circles describe the modern state at 285–330 ppm and stars the strong state at 345–600 ppm; finite critical-shock estimates occur only at 315 and 330 ppm. PlaSim-LSG has no basin-volume estimate. In (e), three-box distances use internal salinity units (multiply by ten for psu); other models use psu. Three-box strength is shown through 896 ppm and resilience through 882 ppm.

Table 2: Default non-dimensional parameters for the Boussinesq model.

Symbol	Value	Physical meaning
Pr	1	Prandtl number (ν/κ_T)
Le	1	Lewis number (κ_T/κ_S)
Ra	4×10^4	Rayleigh number
a	5	Domain length (pole-to-pole)
δ	0.05	Surface boundary-layer thickness
τ_T	0.1	Thermal restoring time scale
τ_S	1.0	Salinity flux time scale
β_s	0.1	Spin-up salinity hemispheric asymmetry
M, N	20, 40	Grid intervals (21×41 points)
Δt	10^{-3}	Time step